GPU considerations

NetApp Solutions

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GPU considerations

GPUs are typically used for graphic visualization (rendering) by performing repetitive arithmetic calculations. This repetitive compute capability is often used for AI and deep learning use cases.

For graphic intensive applications, Microsoft Azure offers the NV series based on the NVIDIA Tesla M60 card with one to four GPUs per VM. Each NVIDIA Tesla M60 card includes two Maxwell-based GPUs, each with 8GB of GDDR5 memory for a total of 16GB.

An NVIDIA license is included with the NV series.

TechPowerUp GPU-Z 2.36.0

Graphics Card | Sensors | Advanced | Validation

Name	NVIDIA Tesla M60		Lookup
GPU	GM204	Revision	FF
Technology	28 nm	Die Size	398 mm²
Release Date	Aug 30, 2015	Transistors	5200M
BIOS Version	84.04.85.00.03		UEFI
Subvendor	NVIDIA	Device ID	10DE 13F2 - 10DE 115E
ROPs/TMUs	64 / 128	Bus Interface	PCI ?
Shaders	2048 Unified	DirectX Support	12 (12_1)
Pixel Fillrate	75.4 GPixel/s	Texture Fillrate	150.8 GTexel/s
Memory Type	GDDR5 (Hynix)	Bus Width	256 bit
Memory Size	8192 MB	Bandwidth	160.4 GB/s
Driver Version	27.21.14.5257 (NVIDIA 452.57) / 2016		
Driver Date	Oct 22, 2020	Digital Signature	WHQL
GPU Clock	557 MHz	Memory 1253 MHz	Boost 1178 MHz
Default Clock	557 MHz	Memory 1253 MHz	Boost 1178 MHz
NVIDIA SLI	Disabled		

Computing: ☑ OpenCL ☐ CUDA ☑ DirectCompute ☑ DirectML

Technologies: ☑ Vulkan ☐ Ray Tracing ☐ PhysX ☑ OpenGL 4.6

NVIDIA Tesla M60 | Close

With NetApp HCI, the H615C GPU contains three NVIDIA Tesla T4 cards. Each NVIDIA Tesla T4 card has a Touring-based GPU with 16GB of GDDR6 memory. When used in a VMware vSphere environment, virtual machines are able to share the GPU, with each VM having dedicated frame buffer memory. Ray tracing is available with the GPUs on the NetApp HCI H615C to produce realistic images including light reflections. Please note that you need to have an NVIDIA license server with a license for GPU features.

TechPowerUp GPU-Z 2.36.0

Graphics Card | Sensors | Advanced | Validation

Name	NVIDIA GRID T4-8Q		Lookup
GPU	TU104	Revision	A1
Technology	12 nm	Die Size	545 mm²
Release Date	Sep 13, 2018	Transistors	13600M
BIOS Version	0.00.00.00.00		UEFI
Subvendor	NVIDIA	Device ID	10DE 1EB8 - 10DE 130F
ROPs/TMUs	8 / 160	Bus Interface	PCI ?
Shaders	2560 Unified	DirectX Support	12 (12_2)
Pixel Fillrate	4.7 GPixel/s	Texture Fillrate	93.6 GTexel/s
Memory Type	GDDR6	Bus Width	256 bit
Memory Size	8192 MB	Bandwidth	Unknown
Driver Version	27.21.14.5257 (NVIDIA 452.57) / 2016		
Driver Date	Oct 22, 2020	Digital Signature	WHQL
GPU Clock	585 MHz	Memory	0 MHz	Shader	N/A
Default Clock	585 MHz	Memory	0 MHz	Shader	N/A
NVIDIA SLI	Disabled		

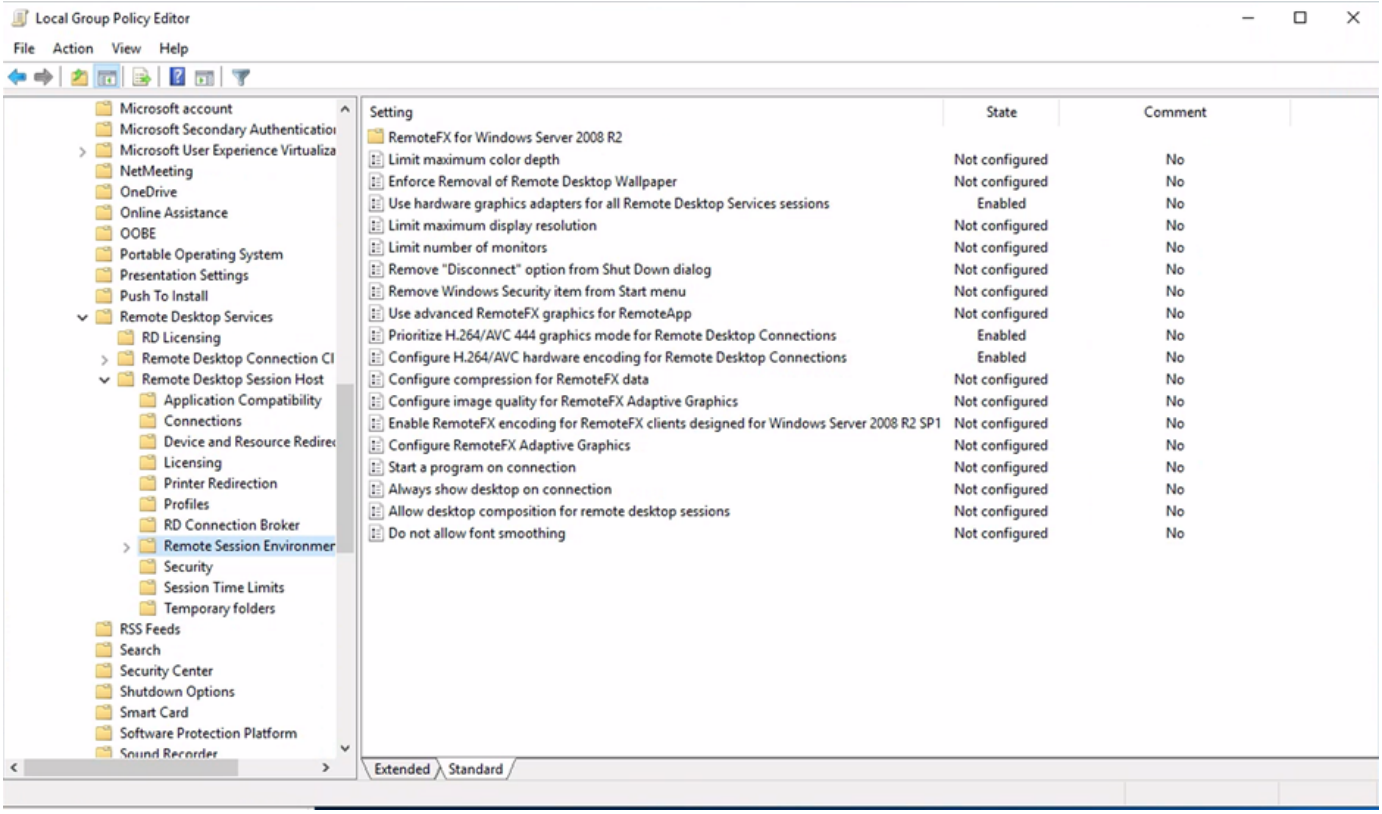
Computing: ☑ OpenCL ☑ CUDA ☑ DirectCompute ☐ DirectML

Technologies: ☑ Vulkan ☑ Ray Tracing ☐ PhysX ☑ OpenGL 4.6

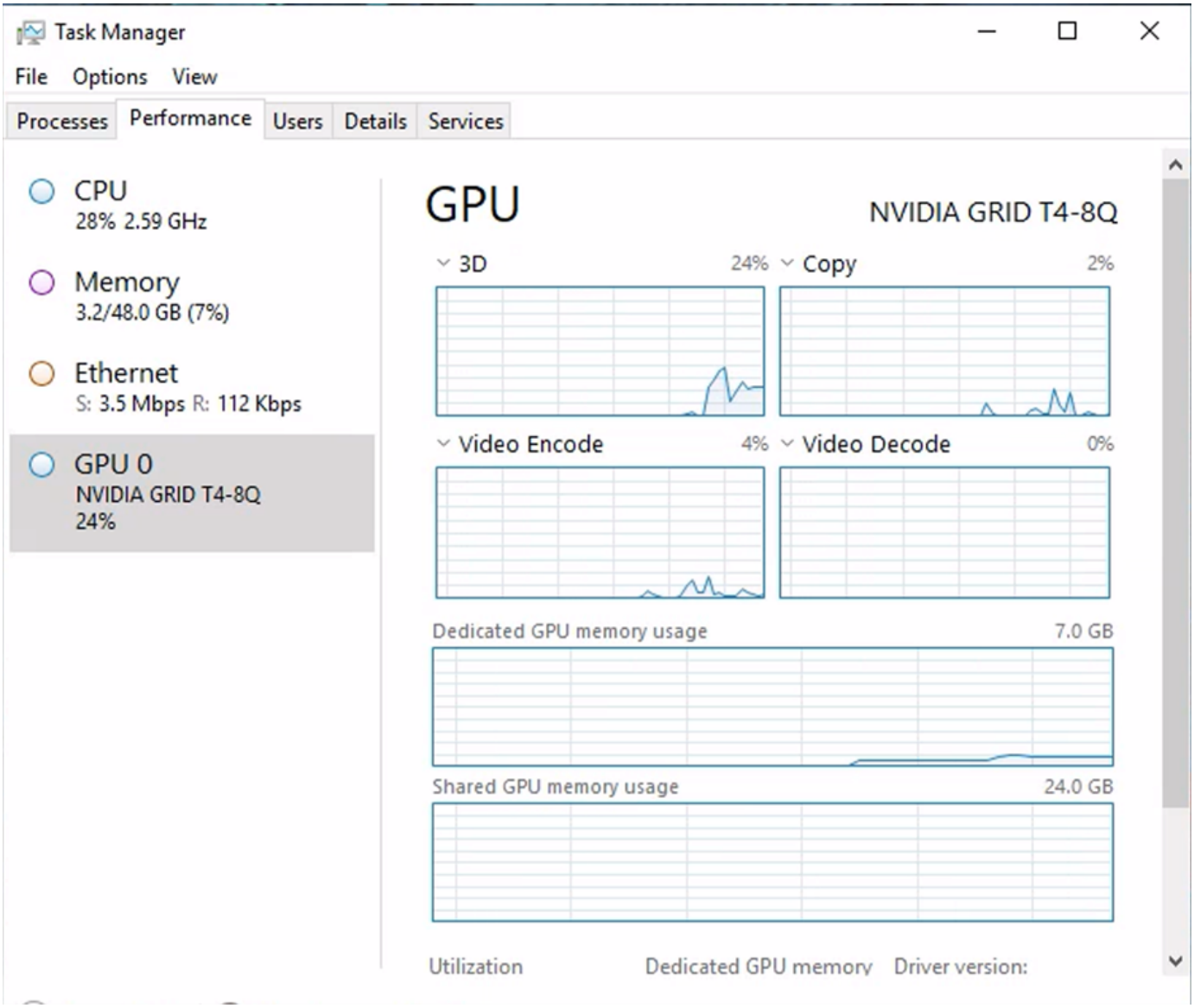
NVIDIA GRID T4-8Q

Close

To use the GPU, you must install the appropriate driver, which can be downloaded from the NVIDIA license portal. In an Azure environment, the NVIDIA driver is available as GPU driver extension. Next, the group policies in the following screenshot must be updated to use GPU hardware for remote desktop service sessions. You should prioritize H.264 graphics mode and enable encoder functionality.



Validate GPU performance monitoring with Task Manager or by using the nvidia-smi CLI when running WebGL samples. Make sure that GPU, memory, and encoder resources are being consumed.



To make sure that the virtual machine is deployed to the NetApp HCI H615C with Virtual Desktop Service, define a site with the vCenter cluster resource that has H615C hosts. The VM template must have the required vGPU profile attached.

For shared multi-session environments, consider allocating multiple homogenous vGPU profiles. However, for high end professional graphics application, it is better to have each VM dedicated to a user to keep VMs isolated.

The GPU processor can be controlled by a QoS policy, and each vGPU profile can have dedicated frame buffers. However, the encoder and decoder are shared for each card. The placement of a vGPU profile on a GPU card is controlled by the vSphere host GPU assignment policy, which can emphasize performance (spread VMs) or consolidation (group VMs).

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